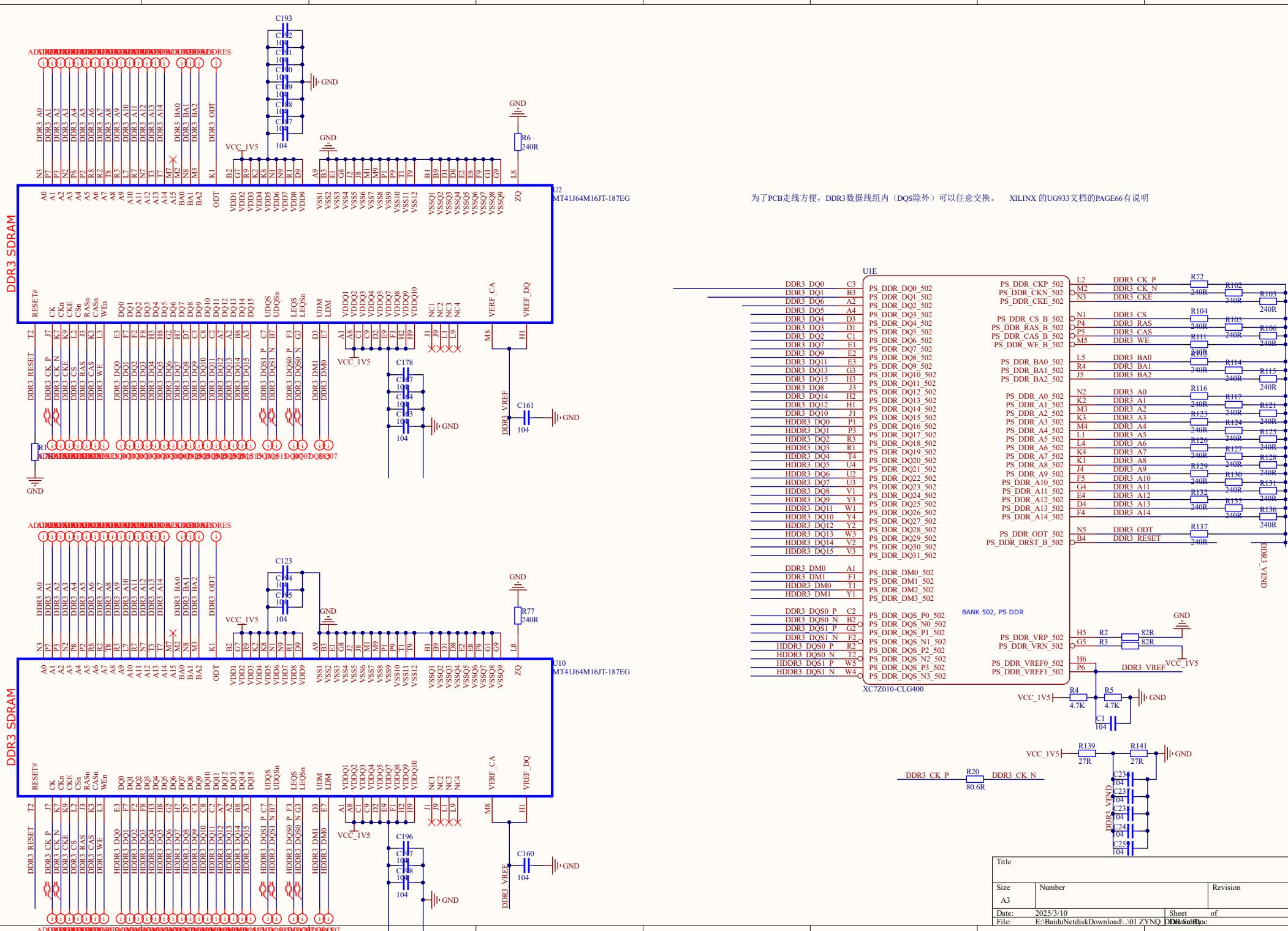
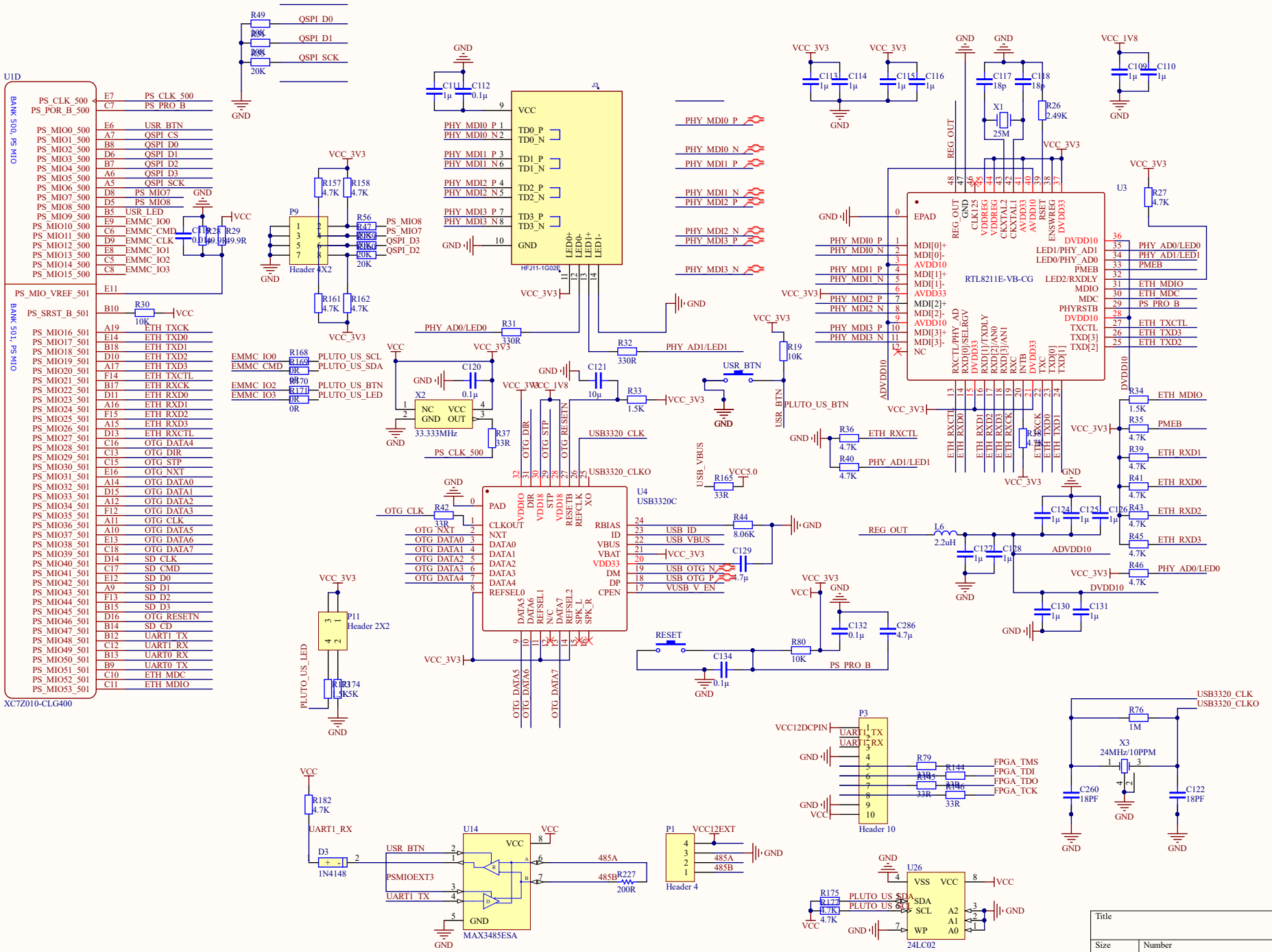
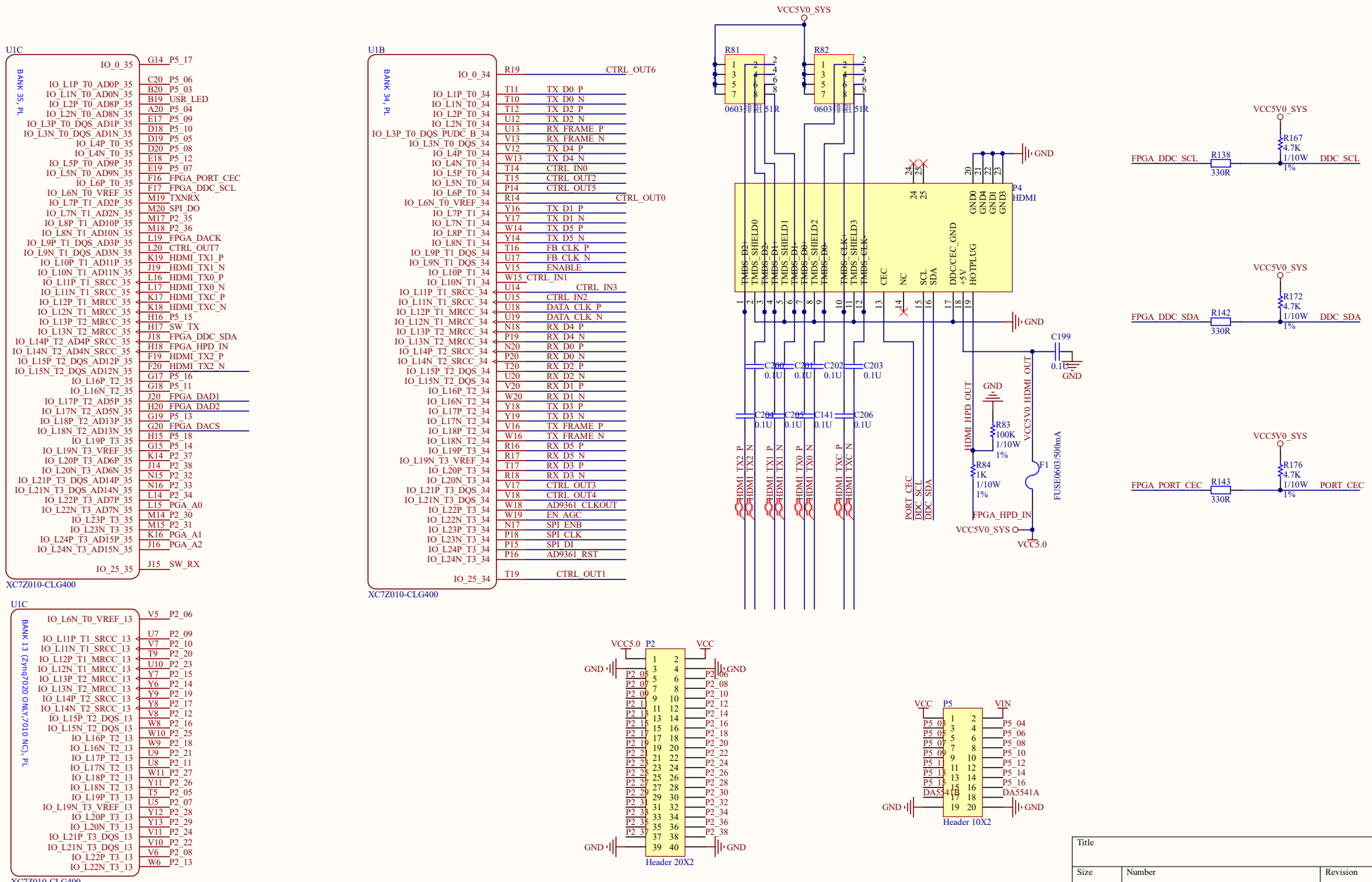
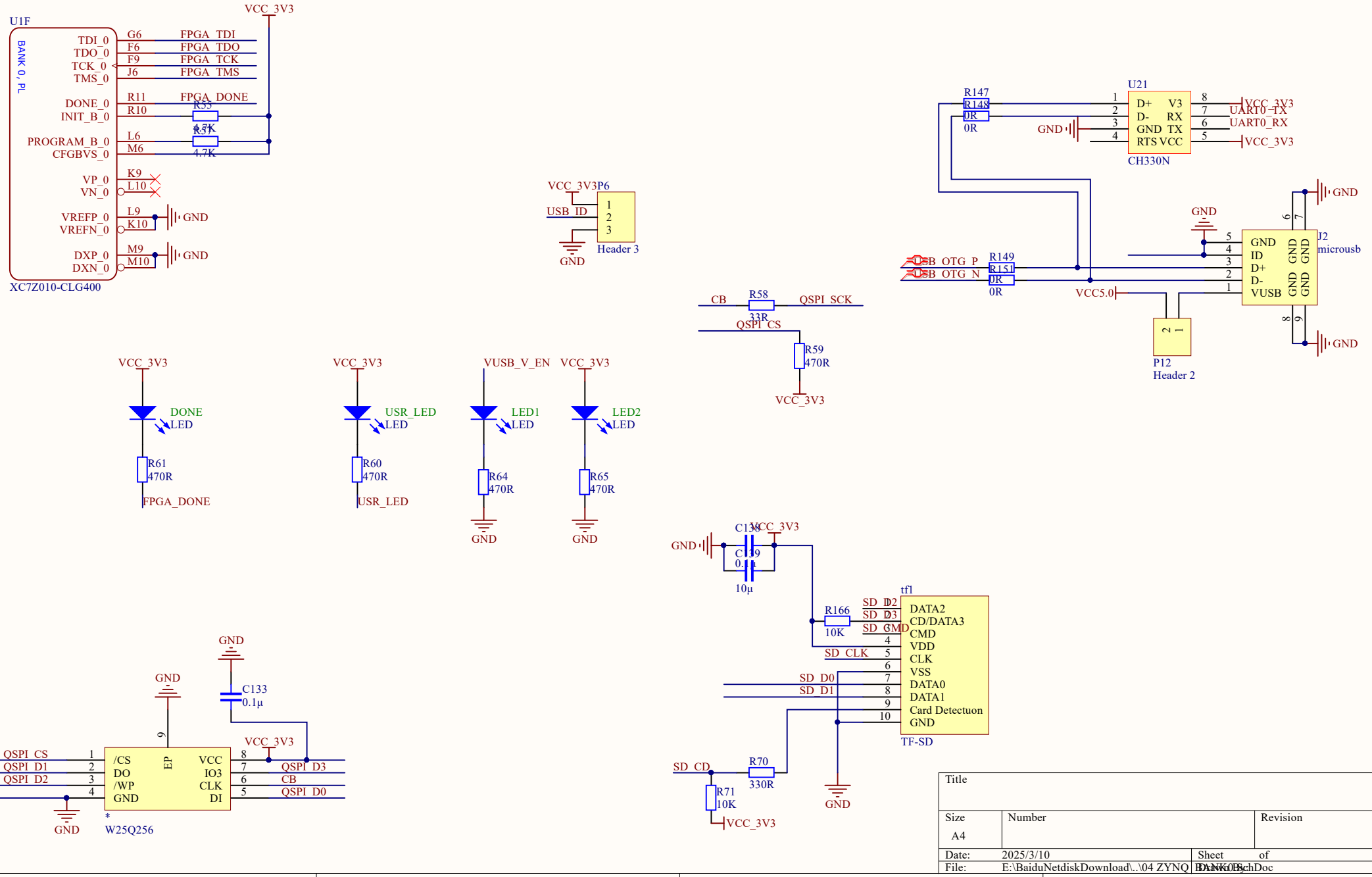
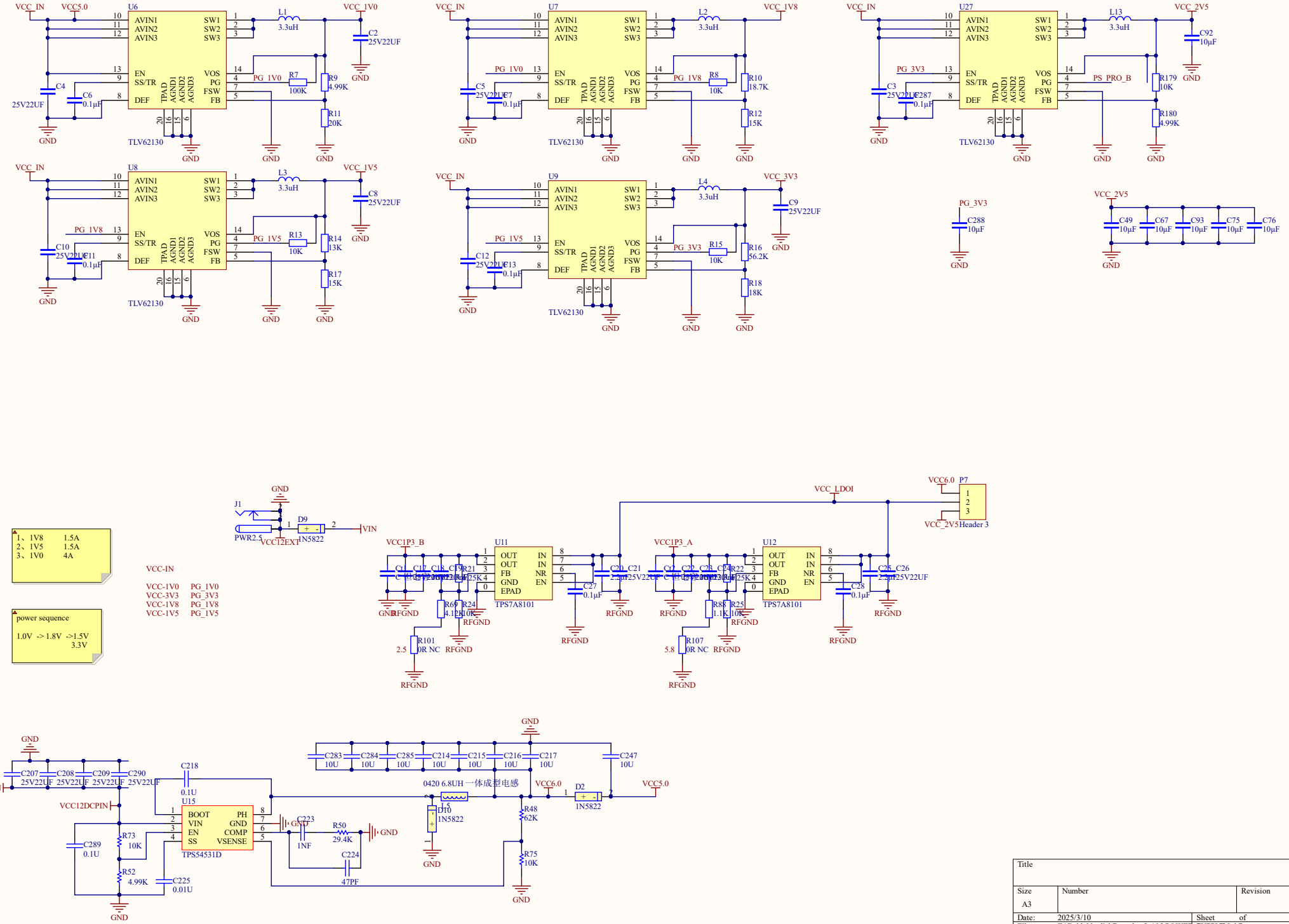
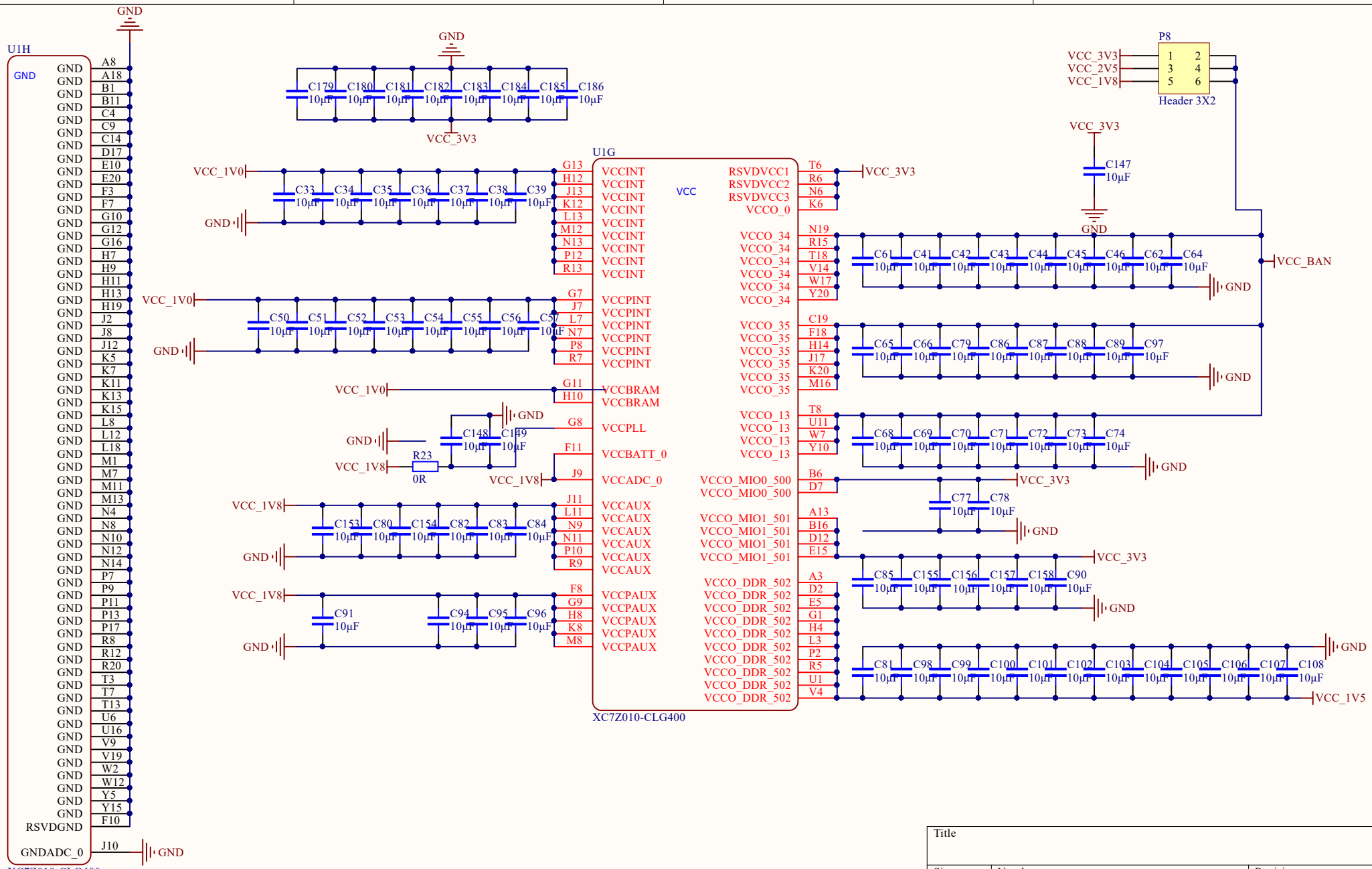

DDR3 SDRAM
U2
MT41J64M16JT-187EG
DDR3 SDRAM
U10
MT41J64M16JT-187EG
为了PCB走线方便，DDR3数据线组内（DQS除外）可以任意交换。
XILINX 的UG933文档的PAGE66有说明
U1E
XC7Z010-CLG400
BANK 502, PS DDR
Title
Size
A3
Number
Revision
Date: 2025/3/10
Sheet of
File: E:\BaiduNetdiskDownload\...\01 ZYNQ_DDR.SchDoc

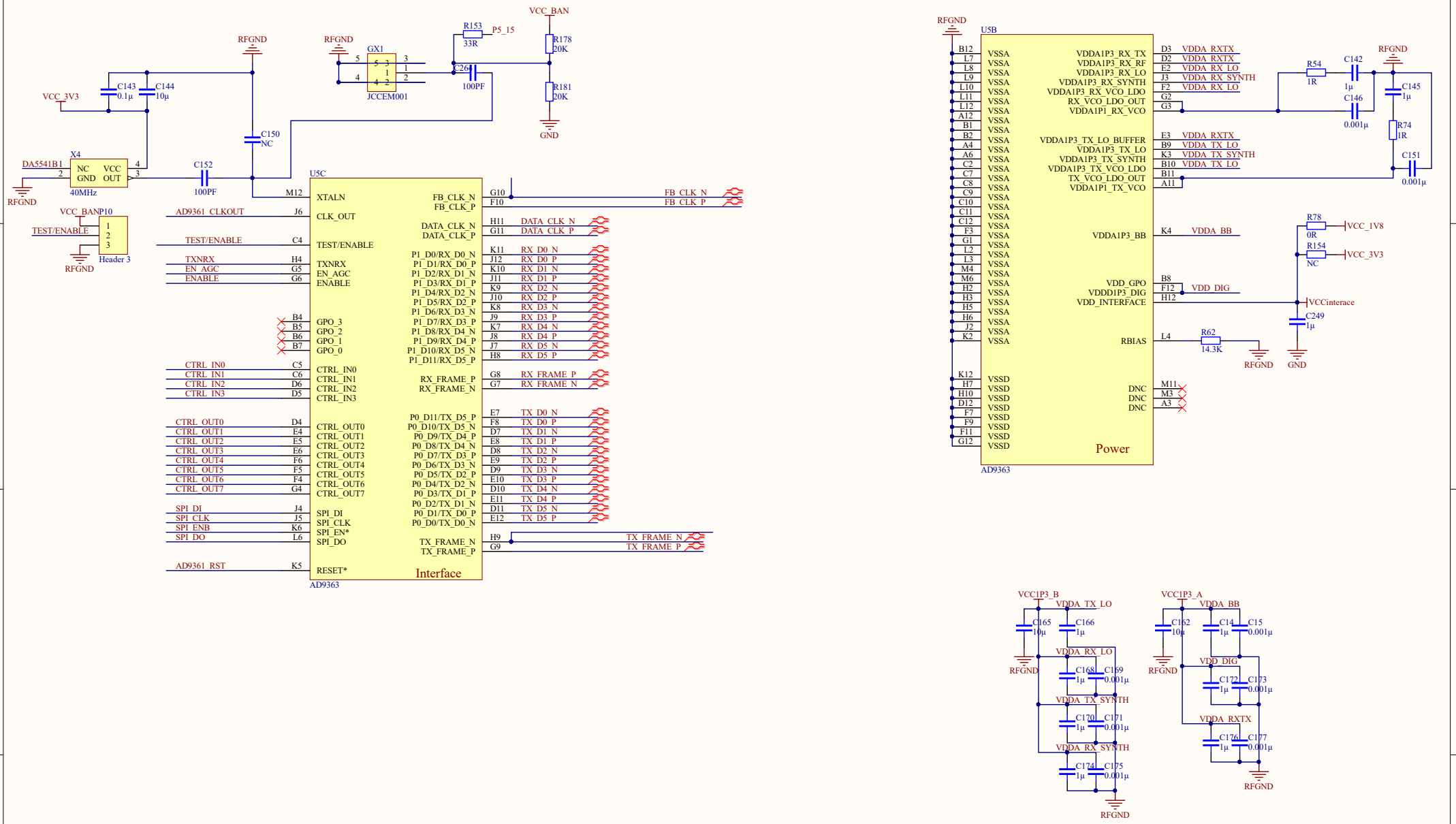

U1D
BANK 500, PS MIO
BANK 501, PS MIO
XC7Z010-CLG400
PS_CLK_500
PS_POR_B_500
PS_MIO0_500
PS_MIO1_500
PS_MIO2_500
PS_MIO3_500
PS_MIO4_500
PS_MIO5_500
PS_MIO6_500
PS_MIO7_500
PS_MIO8_500
PS_MIO9_500
PS_MIO10_500
PS_MIO11_500
PS_MIO12_500
PS_MIO13_500
PS_MIO14_500
PS_MIO15_500
PS_MIO_VREF_501
PS_SRST_B_501
PS_MIO16_501
PS_MIO17_501
PS_MIO18_501
PS_MIO19_501
PS_MIO20_501
PS_MIO21_501
PS_MIO22_501
PS_MIO23_501
PS_MIO24_501
PS_MIO25_501
PS_MIO26_501
PS_MIO27_501
PS_MIO28_501
PS_MIO29_501
PS_MIO30_501
PS_MIO31_501
PS_MIO32_501
PS_MIO33_501
PS_MIO34_501
PS_MIO35_501
PS_MIO36_501
PS_MIO37_501
PS_MIO38_501
PS_MIO39_501
PS_MIO40_501
PS_MIO41_501
PS_MIO42_501
PS_MIO43_501
PS_MIO44_501
PS_MIO45_501
PS_MIO46_501
PS_MIO47_501
PS_MIO48_501
PS_MIO49_501
PS_MIO50_501
PS_MIO51_501
PS_MIO52_501
PS_MIO53_501
QSPI_D0
QSPI_D1
QSPI_SCK
QSPI_D3
QSPI_D2
USR_BTN
USR_LED
EMMC_IO0
EMMC_CMD
EMMC_CLK
EMMC_IO1
EMMC_IO2
EMMC_IO3
ETH_TXCK
ETH_TXD0
ETH_TXD1
ETH_TXD2
ETH_TXD3
ETH_TXCTL
ETH_RXCK
ETH_RXD0
ETH_RXD1
ETH_RXD2
ETH_RXD3
ETH_RXCTL
OTG_DATA4
OTG_DIR
OTG_STP
OTG_NXT
OTG_DATA0
OTG_DATA1
OTG_DATA2
OTG_DATA3
OTG_CLK
OTG_DATA5
OTG_DATA6
OTG_DATA7
SD_CLK
SD_CMD
SD_D0
SD_D1
SD_D2
SD_D3
OTG_RESETN
SD_CD
UART1_TX
UART1_RX
UART0_RX
UART0_TX
ETH_MDC
ETH_MDIO
PS_CLK_500
PS_POR_B
PS_PRO_B
PS_SRST_B_501
PS_MIO_VREF_501
PS_MIO8
PS_MIO7
QSPI_D2
QSPI_D3
PLUTO_US_SCL
PLUTO_US_SDA
PLUTO_US_BTN
PLUTO_US_LED
PHY_MDI0_P
PHY_MDI0_N
PHY_MDI1_P
PHY_MDI1_N
PHY_MDI2_P
PHY_MDI2_N
PHY_MDI3_P
PHY_MDI3_N
PHY_AD0/LED0
PHY_AD1/LED1
USB3320_CLK
USB3320_CLKO
USB_VBUS
USB_ID
USB_VBUS
USB_OTG_N
USB_OTG_P
VUSB_V_EN
RESET
FPGA_TMS
FPGA_TDI
FPGA_TDO
FPGA_TCK
485A
485B
VCC
VCC_3V3
VCC_1V8
VCC5.0
GND
Header 4X2
Header 2X2
Header 10
Header 4
HFJ11-1G02E
RTL8211E-VB-CG
USB3320C
MAX3485ESA
24LC02
1N4148
33.333MHz
24MHz/10PPM
25M
Title
Size
A3
Number
Revision
Date:
2025/3/10
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of
File:
E:\BaiduNetdiskDownload\...\02 ZYNQ PS BANK500-501.SchDoc


U1C
BANK 35, PL
IO_0_35
IO_L1P_T0_AD0P_35
IO_L1N_T0_AD0N_35
IO_L2P_T0_AD8P_35
IO_L2N_T0_AD8N_35
IO_L3P_T0_DQS_AD1P_35
IO_L3N_T0_DQS_AD1N_35
IO_L4P_T0_35
IO_L4N_T0_35
IO_L5P_T0_AD9P_35
IO_L5N_T0_AD9N_35
IO_L6P_T0_35
IO_L6N_T0_VREF_35
IO_L7P_T1_AD2P_35
IO_L7N_T1_AD2N_35
IO_L8P_T1_AD10P_35
IO_L8N_T1_AD10N_35
IO_L9P_T1_DQS_AD3P_35
IO_L9N_T1_DQS_AD3N_35
IO_L10P_T1_AD11P_35
IO_L10N_T1_AD11N_35
IO_L11P_T1_SRCC_35
IO_L11N_T1_SRCC_35
IO_L12P_T1_MRCC_35
IO_L12N_T1_MRCC_35
IO_L13P_T2_MRCC_35
IO_L13N_T2_MRCC_35
IO_L14P_T2_AD4P_SRCC_35
IO_L14N_T2_AD4N_SRCC_35
IO_L15P_T2_DQS_AD12P_35
IO_L15N_T2_DQS_AD12N_35
IO_L16P_T2_35
IO_L16N_T2_35
IO_L17P_T2_AD5P_35
IO_L17N_T2_AD5N_35
IO_L18P_T2_AD13P_35
IO_L18N_T2_AD13N_35
IO_L19P_T3_35
IO_L19N_T3_VREF_35
IO_L20P_T3_AD6P_35
IO_L20N_T3_AD6N_35
IO_L21P_T3_DQS_AD14P_35
IO_L21N_T3_DQS_AD14N_35
IO_L22P_T3_AD7P_35
IO_L22N_T3_AD7N_35
IO_L23P_T3_35
IO_L23N_T3_35
IO_L24P_T3_AD15P_35
IO_L24N_T3_AD15N_35
IO_25_35
G14_P5_17
C20_P5_06
B20_P5_03
B19_USR_LED
A20_P5_04
E17_P5_09
D18_P5_10
D19_P5_05
D20_P5_08
E18_P5_12
E19_P5_07
F16_FPGA_PORT_CEC
F17_FPGA_DDC_SCL
M19_TXNRX
M20_SPI_DO
M17_P2_35
M18_P2_36
L19_FPGA_DACK
L20_CTRL_OUT7
K19_HDMI_TX1_P
J19_HDMI_TX1_N
L16_HDMI_TX0_P
L17_HDMI_TX0_N
K17_HDMI_TXC_P
K18_HDMI_TXC_N
H16_P5_15
H17_SW_TX
J18_FPGA_DDC_SDA
H18_FPGA_HPD_IN
F19_HDMI_TX2_P
F20_HDMI_TX2_N
G17_P5_16
G18_P5_11
J20_FPGA_DAD1
H20_FPGA_DAD2
G19_P5_13
G20_FPGA_DACS
H15_P5_18
G15_P5_14
K14_P2_37
J14_P2_38
N15_P2_32
N16_P2_33
L14_P2_34
L15_PGA_A0
M14_P2_30
M15_P2_31
K16_PGA_A1
J16_PGA_A2
J15_SW_RX
XC7Z010-CLG400
U1B
BANK 34, PL
IO_0_34
IO_L1P_T0_34
IO_L1N_T0_34
IO_L2P_T0_34
IO_L2N_T0_34
IO_L3P_T0_DQS_PUDC_B_34
IO_L3N_T0_DQS_34
IO_L4P_T0_34
IO_L4N_T0_34
IO_L5P_T0_34
IO_L5N_T0_34
IO_L6P_T0_34
IO_L6N_T0_VREF_34
IO_L7P_T1_34
IO_L7N_T1_34
IO_L8P_T1_34
IO_L8N_T1_34
IO_L9P_T1_DQS_34
IO_L9N_T1_DQS_34
IO_L10P_T1_34
IO_L10N_T1_34
IO_L11P_T1_SRCC_34
IO_L11N_T1_SRCC_34
IO_L12P_T1_MRCC_34
IO_L12N_T1_MRCC_34
IO_L13P_T2_MRCC_34
IO_L13N_T2_MRCC_34
IO_L14P_T2_SRCC_34
IO_L14N_T2_SRCC_34
IO_L15P_T2_DQS_34
IO_L15N_T2_DQS_34
IO_L16P_T2_34
IO_L16N_T2_34
IO_L17P_T2_34
IO_L17N_T2_34
IO_L18P_T2_34
IO_L18N_T2_34
IO_L19P_T3_34
IO_L19N_T3_VREF_34
IO_L20P_T3_34
IO_L20N_T3_34
IO_L21P_T3_DQS_34
IO_L21N_T3_DQS_34
IO_L22P_T3_34
IO_L22N_T3_34
IO_L23P_T3_34
IO_L23N_T3_34
IO_L24P_T3_34
IO_L24N_T3_34
IO_25_34
R19 CTRL_OUT6
T11 TX_D0_P
T10 TX_D0_N
T12 TX_D2_P
U12 TX_D2_N
U13 RX_FRAME_P
V13 RX_FRAME_N
V12 TX_D4_P
W13 TX_D4_N
T14 CTRL_IN0
T15 CTRL_OUT2
P14 CTRL_OUT5
R14 CTRL_OUT0
Y16 TX_D1_P
Y17 TX_D1_N
W14 TX_D5_P
Y14 TX_D5_N
T16 FB_CLK_P
U17 FB_CLK_N
V15 ENABLE
W15 CTRL_IN1
U14 CTRL_IN3
U15 CTRL_IN2
U18 DATA_CLK_P
U19 DATA_CLK_N
N18 RX_D4_P
P19 RX_D4_N
N20 RX_D0_P
P20 RX_D0_N
T20 RX_D2_P
U20 RX_D2_N
V20 RX_D1_P
W20 RX_D1_N
Y18 TX_D3_P
Y19 TX_D3_N
V16 TX_FRAME_P
W16 TX_FRAME_N
R16 RX_D5_P
R17 RX_D5_N
T17 RX_D3_P
R18 RX_D3_N
V17 CTRL_OUT3
V18 CTRL_OUT4
W18 AD9361_CLKOUT
W19 EN_AGC
N17 SPI_ENB
P18 SPI_CLK
P15 SPI_DI
P16 AD9361_RST
T19 CTRL_OUT1
XC7Z010-CLG400
U1C
BANK 13 (Zynq7020 ONLY,7010 NC), PL
IO_L6N_T0_VREF_13
IO_L11P_T1_SRCC_13
IO_L11N_T1_SRCC_13
IO_L12P_T1_MRCC_13
IO_L12N_T1_MRCC_13
IO_L13P_T2_MRCC_13
IO_L13N_T2_MRCC_13
IO_L14P_T2_SRCC_13
IO_L14N_T2_SRCC_13
IO_L15P_T2_DQS_13
IO_L15N_T2_DQS_13
IO_L16P_T2_13
IO_L16N_T2_13
IO_L17P_T2_13
IO_L17N_T2_13
IO_L18P_T2_13
IO_L18N_T2_13
IO_L19P_T3_13
IO_L19N_T3_VREF_13
IO_L20P_T3_13
IO_L20N_T3_13
IO_L21P_T3_DQS_13
IO_L21N_T3_DQS_13
IO_L22P_T3_13
IO_L22N_T3_13
V5_P2_06
U7_P2_09
V7_P2_10
T9_P2_20
U10_P2_23
Y7_P2_15
Y6_P2_14
Y9_P2_19
Y8_P2_17
V8_P2_12
W8_P2_16
W10_P2_25
W9_P2_18
U9_P2_21
U8_P2_11
W11_P2_27
Y11_P2_26
T5_P2_05
U5_P2_07
Y12_P2_28
Y13_P2_29
V11_P2_24
V10_P2_22
V6_P2_08
W6_P2_13
XC7Z010-CLG400
VCC5V0_SYS
R81
R82
0603
51R
P4
HDMI
TMDS_SHIELD0
TMDS_SHIELD1
TMDS_SHIELD2
TMDS_SHIELD3
CEC
NC
SCL
SDA
DDC/CEC_GND
+5V
HOTPLUG
GND0
GND4
GND1
GND3
GND
C199
0.1U
C206
C201
C202
C203
C204
C205
C141
C206
HDMI_TX2_P
HDMI_TX2_N
HDMI_TX1_P
HDMI_TX1_N
HDMI_TX0_P
HDMI_TX0_N
HDMI_TXC_P
HDMI_TXC_N
PORT_CEC
DDC_SCL
DDC_SDA
HDMI_HPD_OUT
R83
100K
1/10W
1%
R84
1K
1/10W
1%
FPGA_HPD_IN
VCC5V0_HDMI_OUT
F1
FUSE0603:500mA
VCC5.0
FPGA_DDC_SCL
R138
330R
R167
4.7K
1/10W
1%
DDC_SCL
FPGA_DDC_SDA
R142
R172
DDC_SDA
FPGA_PORT_CEC
R143
R176
PORT_CEC
P2
Header 20X2
P5
VIN
DA5541A
Header 10X2
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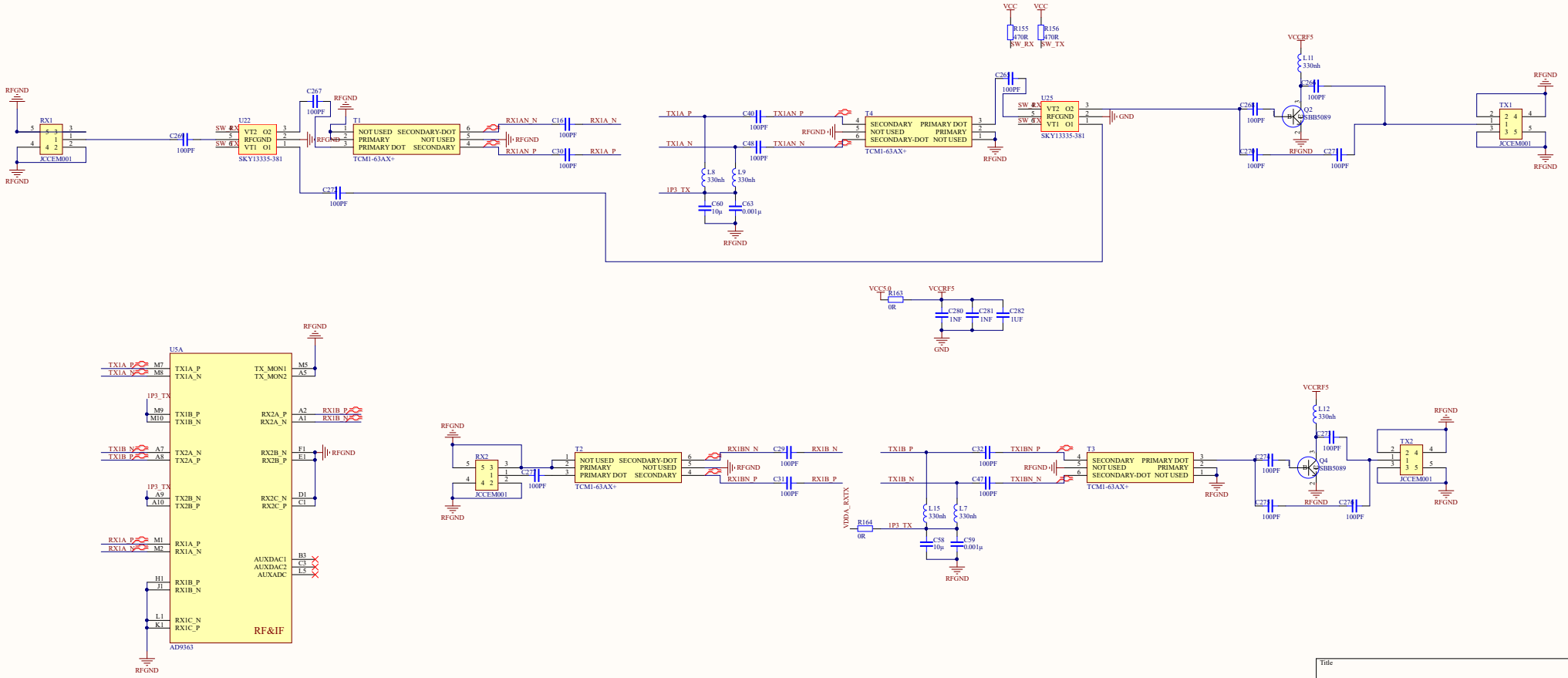

U1F
BANK 0, PL
TDI_0
TDO_0
TCK_0
TMS_0
DONE_0
INIT_B_0
PROGRAM_B_0
CFGBVS_0
VP_0
VN_0
VREFP_0
VREFN_0
DXP_0
DXN_0
G6
F6
F9
J6
R11
R10
L6
M6
K9
L10
L9
K10
M9
M10
FPGA_TDI
FPGA_TDO
FPGA_TCK
FPGA_TMS
FPGA_DONE
R55
4.7K
R57
4.7K
VCC_3V3
GND
XC7Z010-CLG400
VCC_3V3
P6
USB_ID
Header 3
GND
U21
CH330N
D+
D-
GND
RTS
V3
RX
TX
VCC
R147
R148
0R
R149
R151
0R
USB_OTG_P
USB_OTG_N
VCC_3V3
UART0_TX
UART0_RX
VCC5.0
J2
microusb
GND
ID
D+
D-
VUSB
P12
Header 2
CB
R58
33R
QSPI_SCK
QSPI_CS
R59
470R
VCC_3V3
DONE
LED
R61
470R
FPGA_DONE
USR_LED
R60
470R
VUSB_V_EN
LED1
R64
470R
LED2
R65
470R
C138
10µ
C139
0.1µ
tf1
TF-SD
DATA2
CD/DATA3
CMD
VDD
CLK
VSS
DATA0
DATA1
Card Detectuon
R166
10K
SD_D2
SD_D3
SD_CMD
SD_CLK
SD_D0
SD_D1
SD_CD
R70
330R
R71
10K
C133
0.1µ
QSPI_CS
QSPI_D1
QSPI_D2
/CS
DO
/WP
EP
IO3
CLK
DI
QSPI_D3
QSPI_D0
W25Q256
Title
Size
A4
Number
Revision
Date:
2025/3/10
Sheet
of
File:
E:\BaiduNetdiskDownload\..\04 ZYNQ
.SchDoc


VCC_IN
VCC5.0
U6
TLV62130
L1
3.3uH
VCC_1V0
C2
25V22UF
C4
25V22UF
C6
0.1μF
R7
100K
PG_1V0
R9
4.99K
R11
20K
U7
L2
VCC_1V8
R8
10K
R10
18.7K
R12
15K
U27
L13
VCC_2V5
C92
10μF
PG_3V3
PS_PRO_B
R179
10K
R180
4.99K
U8
L3
VCC_1V5
C8
25V22UF
PG_1V8
R13
10K
R14
13K
R17
15K
U9
L4
VCC_3V3
C9
25V22UF
PG_1V5
R15
10K
R16
56.2K
R18
18K
C288
10μF
C49
10μF
C67
10μF
C93
10μF
C75
10μF
C76
10μF
GND
1、1V8 1.5A
2、1V5 1.5A
3、1V0 4A
power sequence
1.0V -> 1.8V ->1.5V
3.3V
VCC-IN
VCC-1V0 PG_1V0
VCC-3V3 PG_3V3
VCC-1V8 PG_1V8
VCC-1V5 PG_1V5
J1
PWR2.5
VCC12EXT
D9
1N5822
VIN
VCC_LDOI
VCC6.0
P7
Header 3
VCC_2V5
VCC1P3_B
VCC1P3_A
U11
U12
TPS7A8101
RFGND
R69
4.12K
R24
10K
R101
0R NC
2.5
R88
1.1K
R25
10K
R107
0R NC
5.8
C283
C284
C285
C214
C215
C216
C217
C247
10U
0420 6.8UH 一体成型电感
D10
1N5822
D2
1N5822
VCC5.0
C207
C208
C209
C290
25V22UF
C218
0.1U
VCC12DCPIN
U15
TPS54531D
BOOT
VIN
EN
SS
PH
GND
COMP
VSENSE
C223
1NF
R50
29.4K
C224
47PF
R48
62K
R75
10K
C289
0.1U
R73
10K
R52
4.99K
C225
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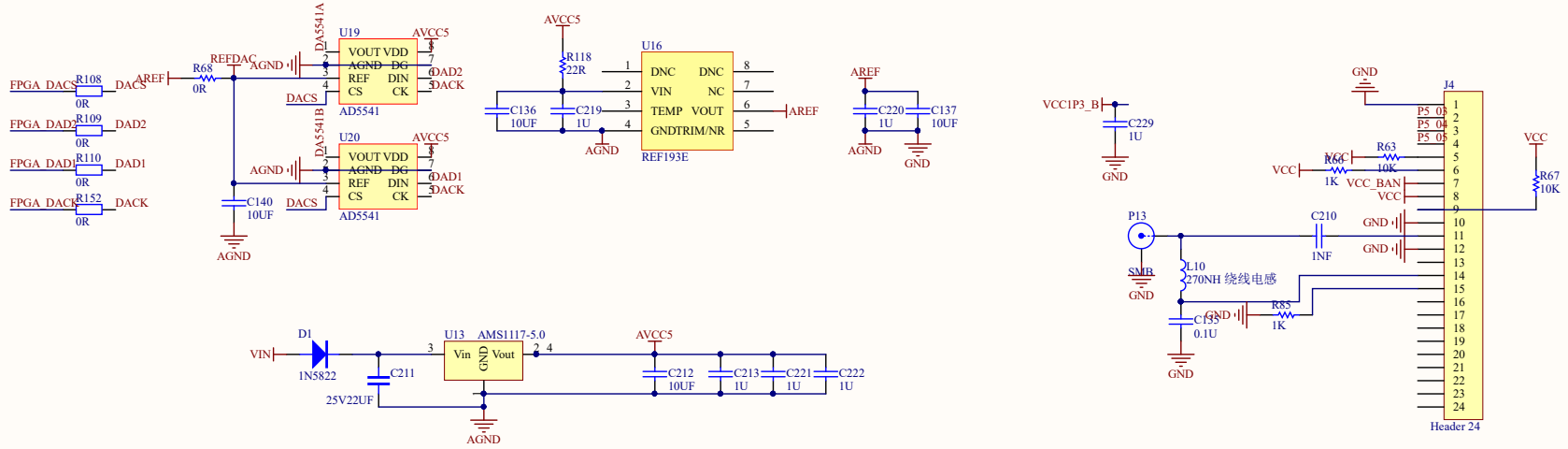

U1H
XC7Z010-CLG400
U1G
XC7Z010-CLG400
P8
Header 3X2
GND
VCC_3V3
VCC_1V0
VCC_1V8
VCC_1V5
VCC_BAN
R23
0R
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Sheet of
SchDoc



Title		
Size A3	Number	Revision
Date: 2025/3/10	Sheet	of
File: E:\BaiduNetdiskDownload\...\07 AD9363_SDRDoc	Drawn By:	



Title		
Size: A2	Number	Revision
Date: 2025/3/10	Sheet of	
File: E:\BaiduNetdiskDownload\...\08 BALUN.SchDoc	Drawn By:	



Title		
Size A3	Number	Revision
Date: 2025/3/10	Sheet of	
File: E:\BaiduNetdiskDownload\...\GPS.SchDoc	Drawn By:	

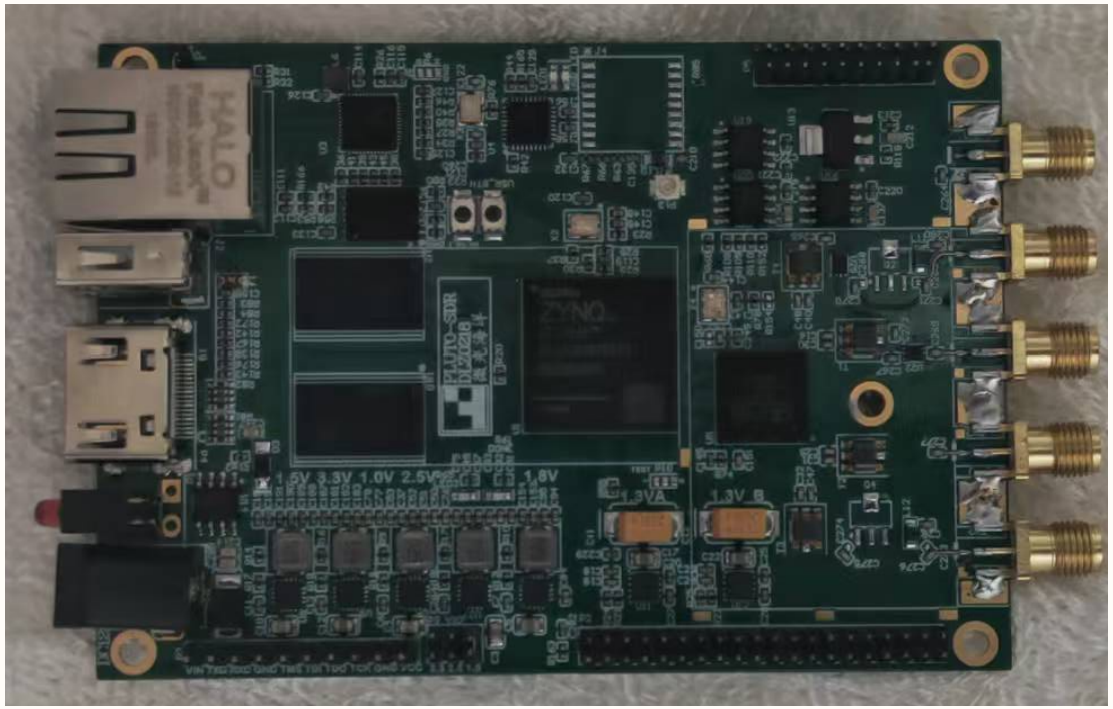
ETH 1000

USB A口

HDMI

pluto 指示灯

DC-12V



REFCLK

TX1

RX1

RX2

TX2

Title		
Size A3	Number	Revision
Date: 2025/3/10	Sheet of	
File: E:\BaiduNetdiskDownload\...接口示意图.SchDoc	Drawn By:	

DC12
J1
P11
J3
P3
VIN TXD RXD GND TMS TDI TDO TCK GND VCC
3.3 2.5 1.8
1.5V 3.3V 1.0V 2.5V
1.8V
1.3VA
1.3V B
PLUTO SDR
DLZ0218
微光海洋
U10
U2
U1
U3
U5
P5
A / I
B / O
TX2
RX2
DDS
RX1
AOUT
TX1
EXT REF
GX1
PCB EDGE
QA/RGS
26JUL16
CNSMARAL687W375H171